

Surface Water Infrastructure Design

ENCN442 – TERM 2 Review



Exam

- 2.5 hours (150 minutes). $\frac{1}{4}$ for each part of the course.
- Water Infrastructure:
 - Channel design (allowable and shear stress)
 - Hydraulic Structure design
 - Stream restoration /Planning / Concepts
- Mixture of conceptual questions and problems

Threshold Channel Design

Table 8-1: General guidance for selecting the most appropriate channel design technique

Technique	Significant sediment load and movable channel boundaries	Boundary material smaller than sand size	Boundary material larger than sand size	Boundary material does not act as discrete particles	No base flow in channel. Climate can support permanent vegetation
Allowable Velocity		X			
Allowable Shear Stress			X		
Tractive Power				X	
Grass Lined / tractive stress					X
Alluvial Channel Design Techniques	X				

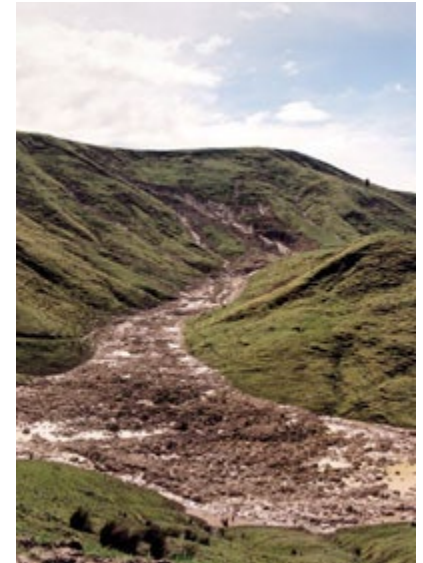
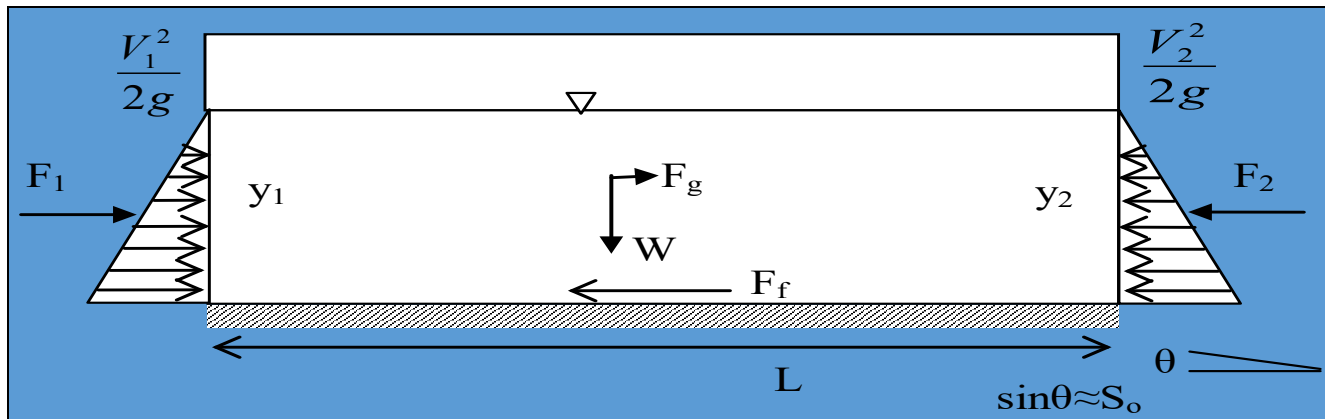
Determine Allowable Velocity

Table 8-3. Maximum Permissible Canal Velocities (Fortier and Scobey, 1926)

Original material excavated for canals	Mean velocity, for straight canals of small slope, after aging with flow depths less than 3 ft (0.9m)					
	Clear water, no detritus		Water transporting colloidal silts		Water transporting noncolloidal silts, sands, gravels or rock fragments	
	ft/sec	m/sec	ft/sec	m/sec	ft/sec	m/sec
Fine Sand noncolloidal	1.5	0.46	2.5	0.76	1.5	0.46
Sandy loam, noncolloidal	1.75	0.53	2.5	0.76	2.0	0.61
Silt loam, noncolloidal	2.0	0.61	3.0	0.91	2.0	0.61
Alluvial silt, noncolloidal	2.0	0.61	3.5	1.07	2.0	0.61
Ordinary firm loam	2.5	0.76	3.5	1.07	2.25	0.69
Volcanic ash	2.5	0.76	3.5	1.07	2.0	0.61
Stiff clay, very colloidal	3.75	1.14	5.0	1.52	3.0	0.91
Alluvial silt, colloidal	3.75	1.14	5.0	1.52	3.0	0.91
Shales and hardpans	6.0	1.83	6.0	1.83	5.0	1.52
Fine gravel	2.5	0.76	5.0	1.52	3.75	1.14
Graded, loam to cobbles when noncolloidal	3.75	1.14	5.0	1.52	5.0	1.52
Graded silt to cobbles when colloidal	4.0	1.22	5.5	1.68	5.0	1.52
Coarse gravel, noncolloidal	4.0	1.22	6.0	1.83	6.5	1.98
Cobbles and shingles	5.0	1.52	5.5	1.68	6.5	1.98

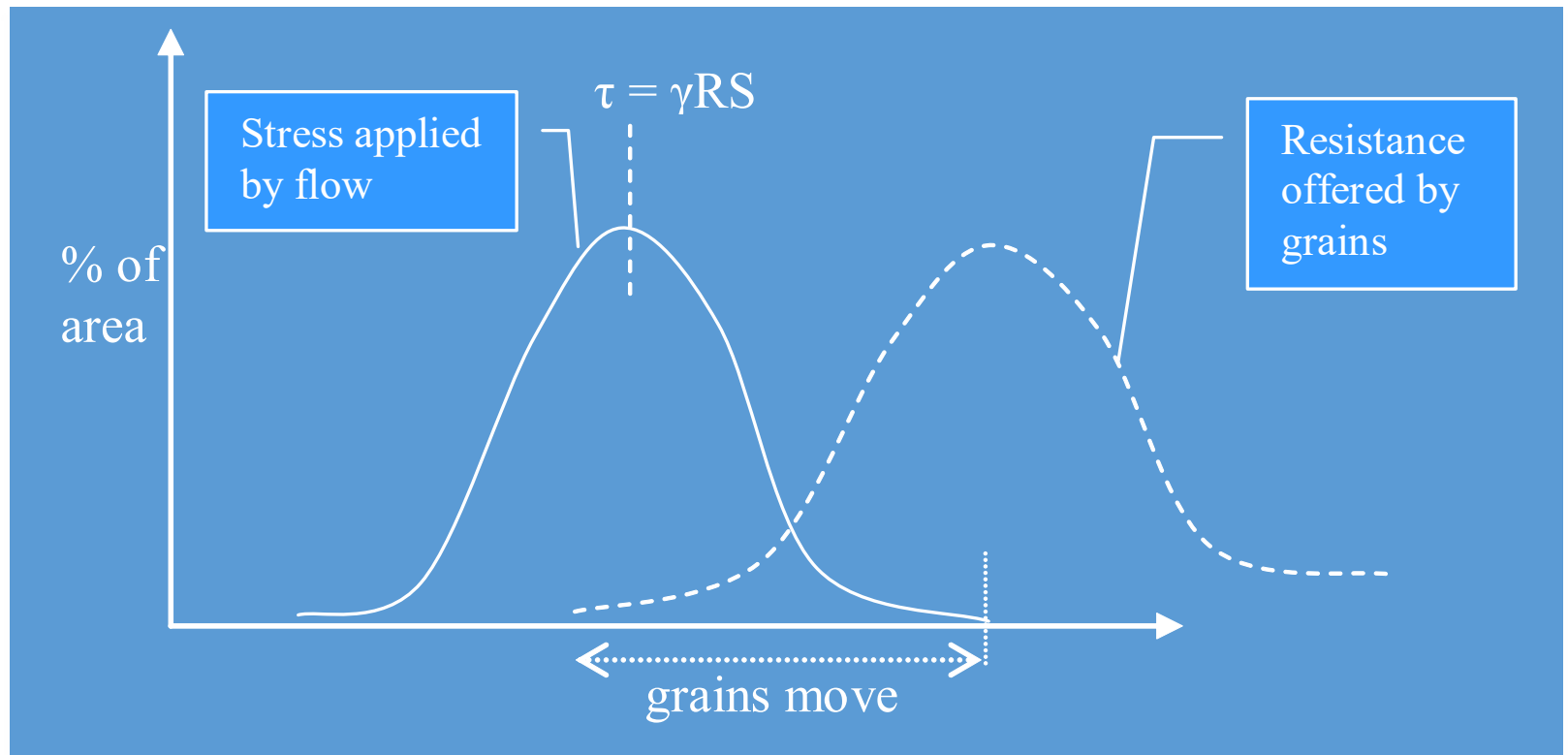
Sediment transport

- Shear stress: $\tau_o = \gamma R S_o$

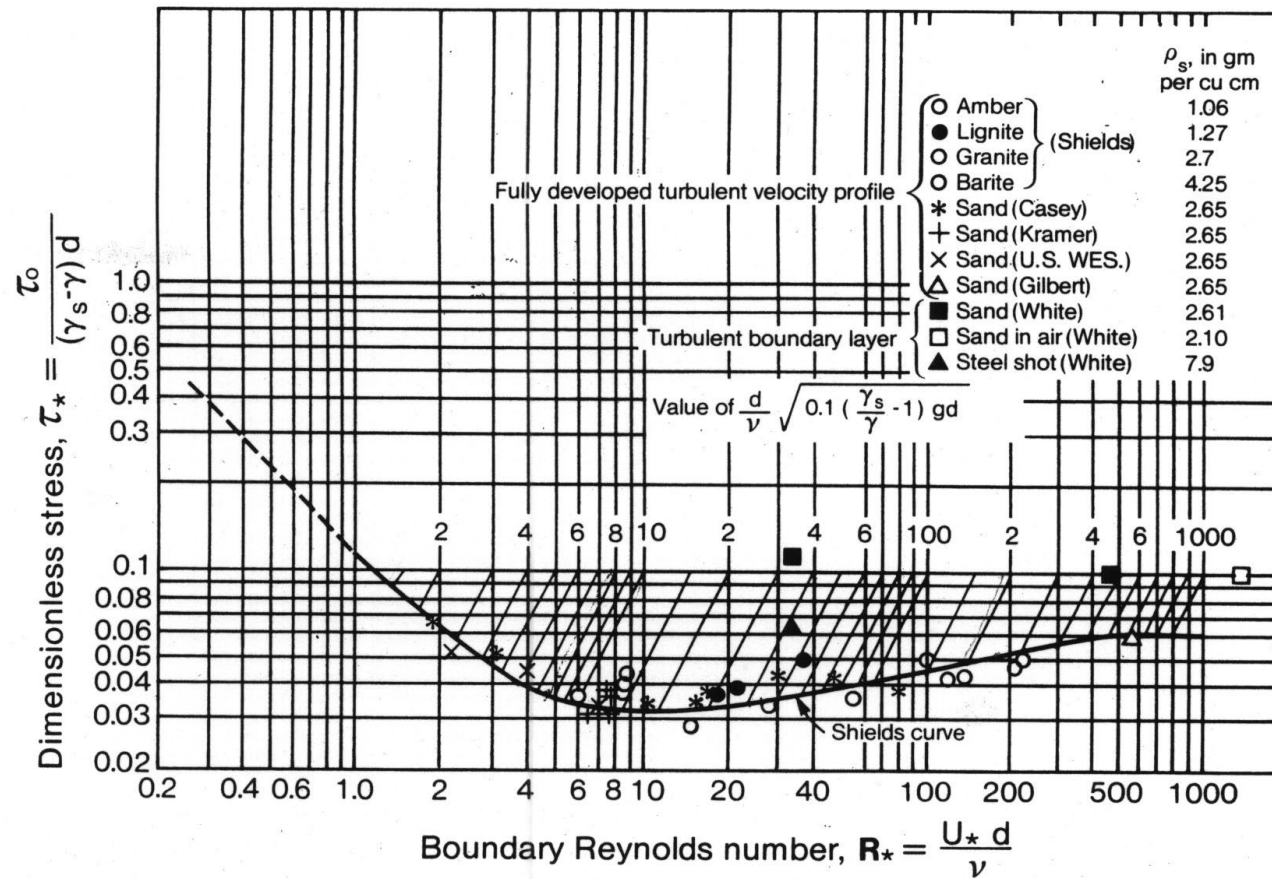


Resisting forces

- Resisting forces: comprise of the weight and any interference forces (mechanical or cohesive) between grains.



Shield's diagram



Shields curve approximation

The dimensionless stress for the Shields curve can be approximated using the following equation:

$$\tau^* = 0.22 \beta + 0.06 \times 10^{-7.7\beta}$$
$$\beta = \left(\frac{1}{\nu} \sqrt{\left(\frac{\gamma_s - \gamma}{\gamma} \right) g d^3} \right)^{-0.6}$$

Note: “d” in meters, $g = 9.8 \text{ m/s}^2$

ν = kinematic viscosity (ie. $1.308 \times 10^{-6} \text{ m}^2/\text{s}$ at ~10 degrees C)

Hydraulic structures

- Design of hydraulic structures
- Identifying a need... energy dissipation, flow measurements, control...
- Drop structures
- Weirs
- Sluice gates
- Chutes
- Culverts (inlet conditions / outlet conditions)

Hydraulic design

- Functional requirements of hydraulic design:
 - Sufficient capacity to pass the design discharge
 - Adequate overflow capacity (in case of extreme events or miss operation)
 - Kinetic energy of the discharge must be dissipated within the confines of the structure
 - Protect structure
 - Protect downstream channel from damage or erosion
 - Minimize hydraulic loss in certain transitions by correct structural proportioning

Using what is available for design

- Equations
- Nomographs....

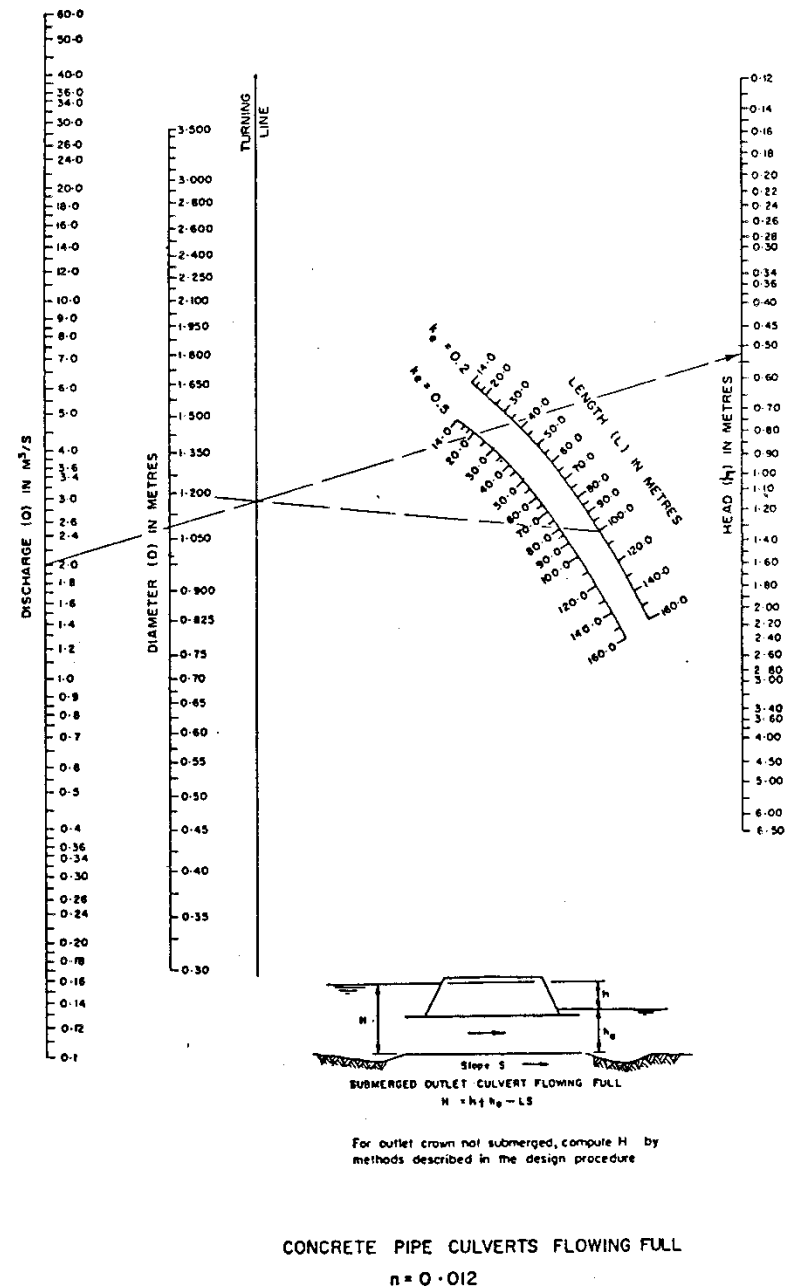


Figure 8: Nomograph for head loss h , under Outlet Control - Circular Culverts

Basic in-channel structures

- Chutes
- Sluice gates
- Culverts



Channel design – other considerations

- Stream / river restoration
- Other factors in design

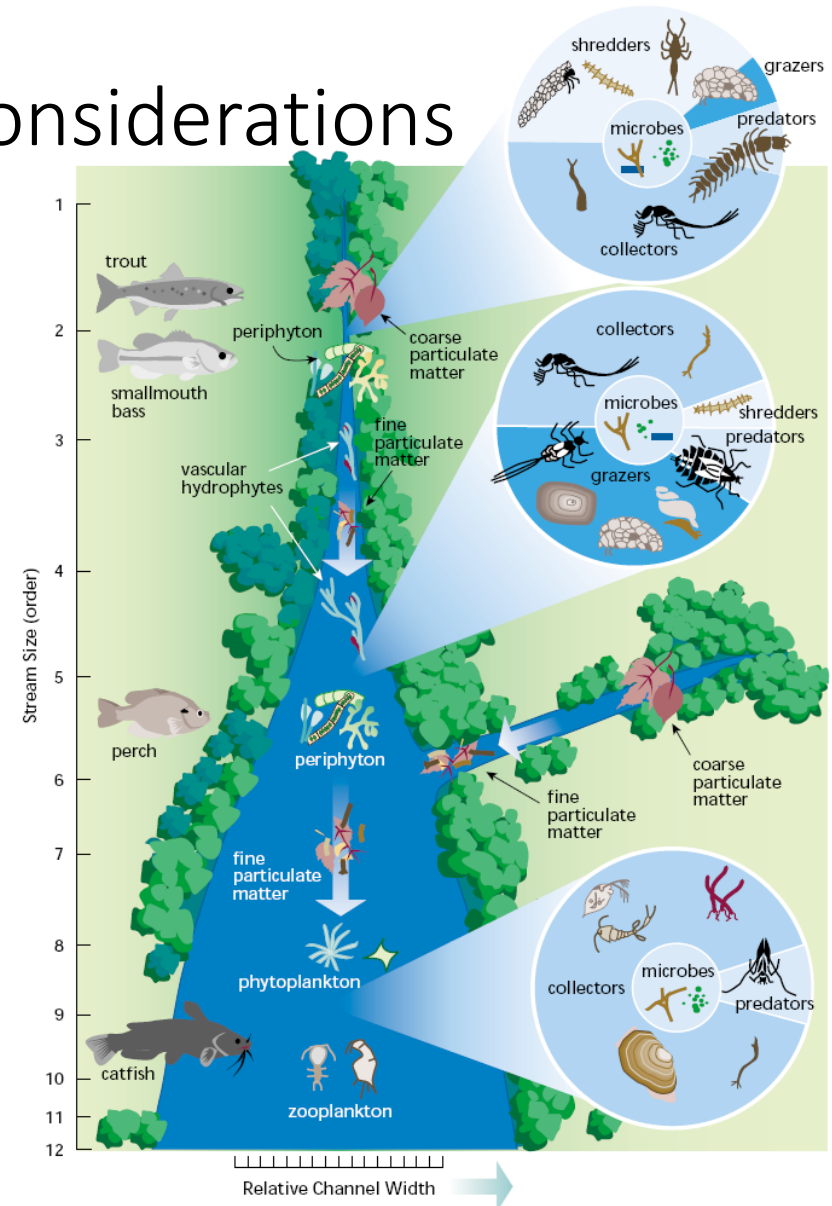
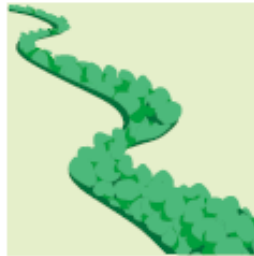


Figure 1.34: The River Continuum Concept. The concept proposes a relationship between stream size and the progressive shift in structural and functional attributes. Source: Vannote et al. (1980). Published with the permission of NRC Research Press.

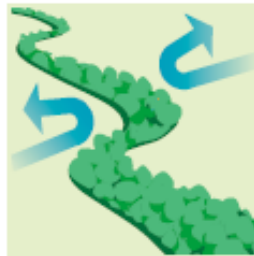
Critical ecosystem functions

Habitat—the spatial structure of the environment which allows species to live, reproduce, feed, and move.



Habitat

Barrier—the stoppage of materials, energy, and organisms.



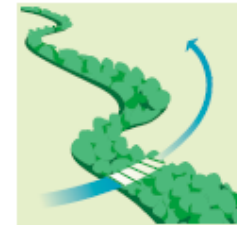
Barrier

Conduit—the ability of the system to transport materials, energy, and organisms.



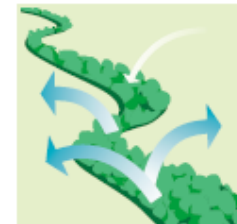
Conduit

Filter—the selective penetration of materials, energy, and organisms.



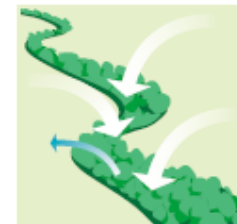
Filter

Source—a setting where the output of materials, energy, and organisms exceeds input.



Source

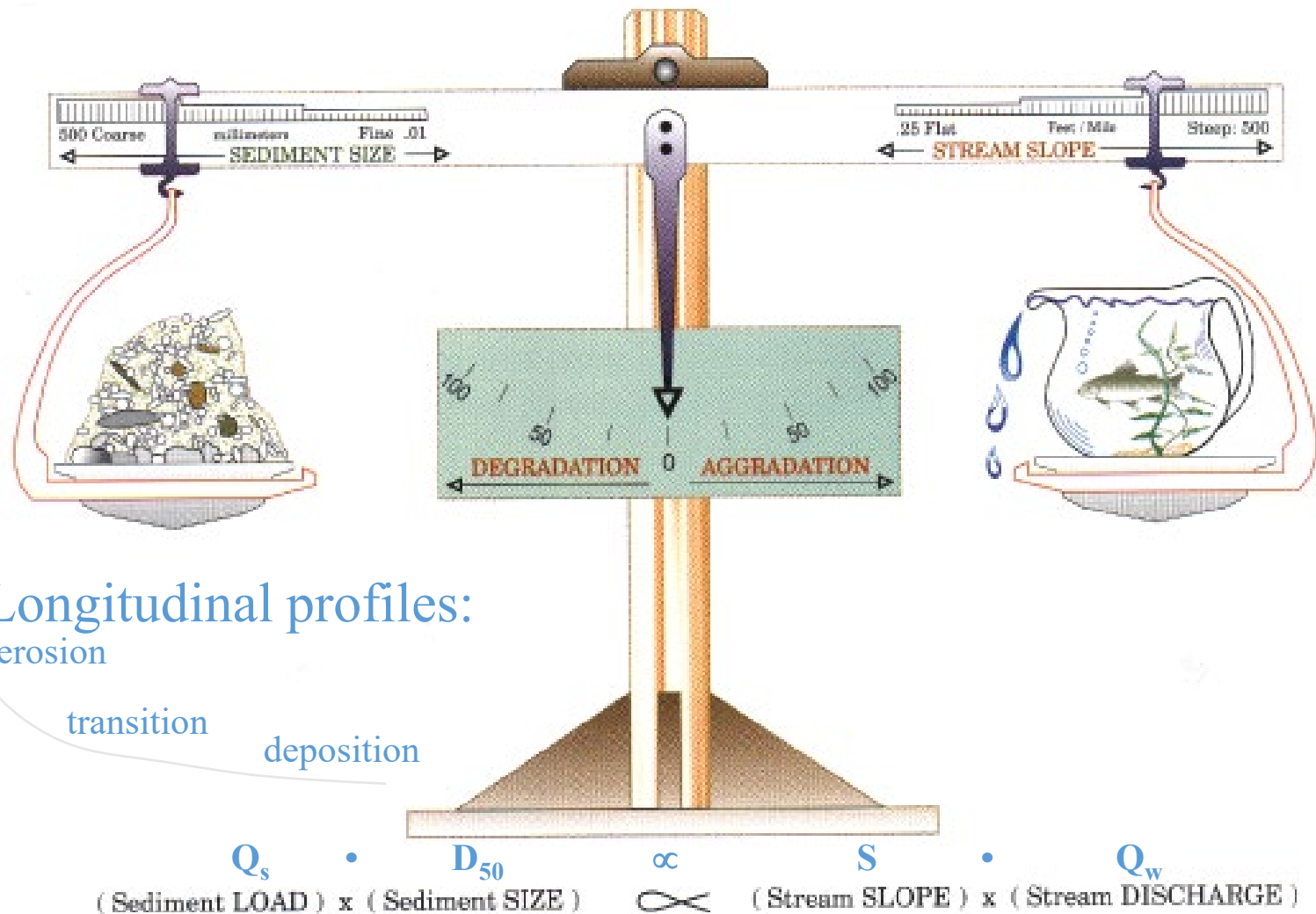
Sink—a setting where the input of water, energy, organisms and materials exceeds output.



Sink

Figure 2.37: Critical ecosystem functions. Six functions can be summarized as a set of basic, common themes recurring in a variety of settings.

Lane's Model



Adaptive Management



- *Modify plans using monitoring, technical, and social feedback*
- *Track restoration policy, programs, and individual projects as feedback for further restoration policy and program redesign*
- *Restoration initiatives: recommend annual assessments*
 - *use monitoring data and other data/expertise*
 - *midcourse corrections or alternative actions*
 - *link reporting/monitoring schedules for midcourse corrections*
- *Manager may contract some/all monitoring, but periodically must visit sites, review reports, discuss with contractors.*